



Biochip Course

Case Study

Impedance Flow Cytometer

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a.y. 2022/2023

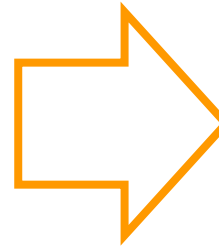
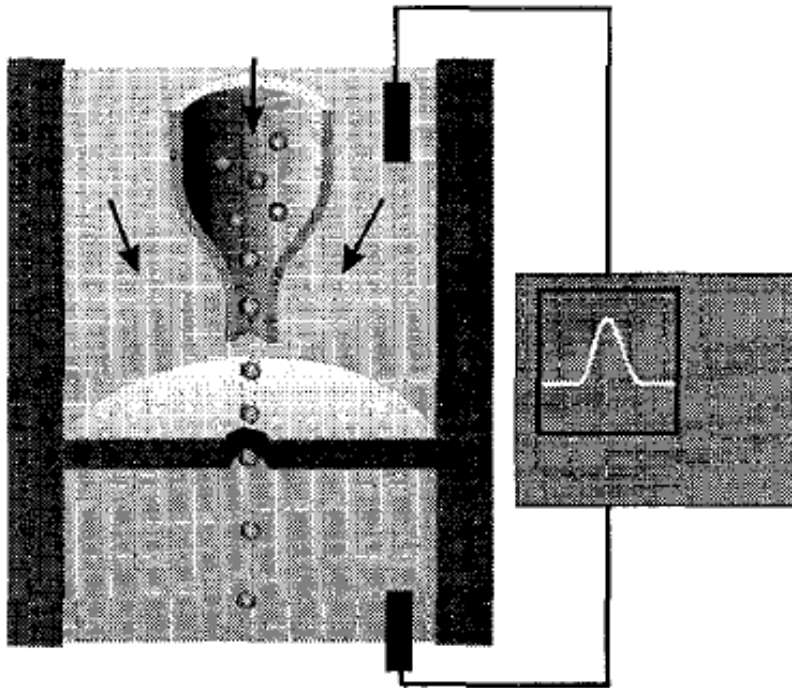
April 26th 2023



The Coulter Counter

Invented by W.H. Coulter 1950

A routine lab. instrument

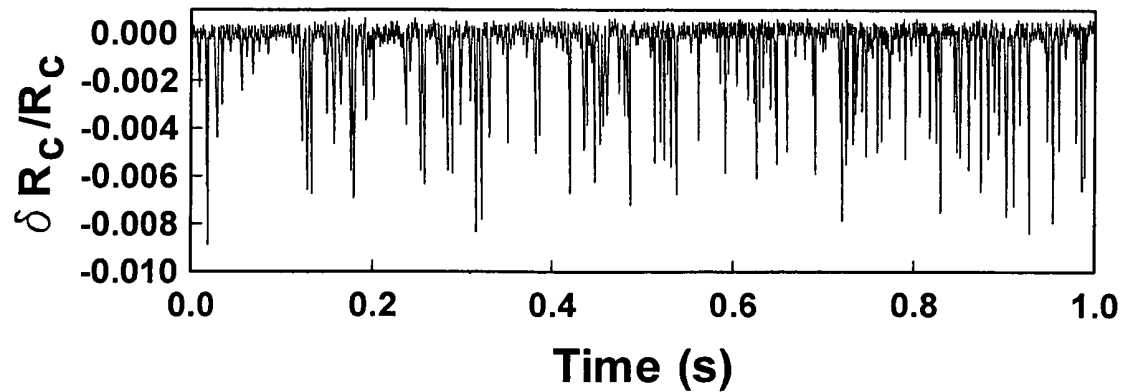


March 24, 1970 W. H. COULTER ET AL 3,502,974
SIGNAL MODULATED APPARATUS FOR GENERATING AND DETECTING
RESISTIVE AND REACTIVE CHANGES IN A MODULATED CURRENT
PATH FOR PARTICLE CLASSIFICATION AND ANALYSIS
Filed May 23, 1966 9 Sheets-Sheet 1

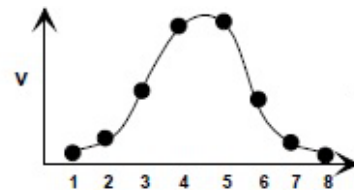
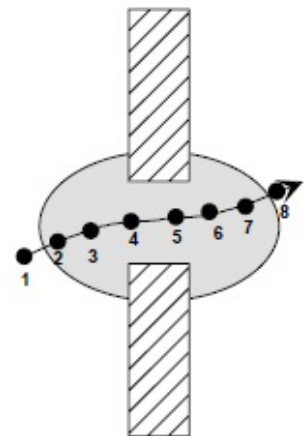
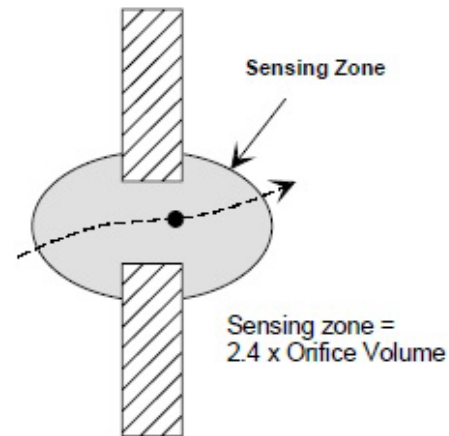
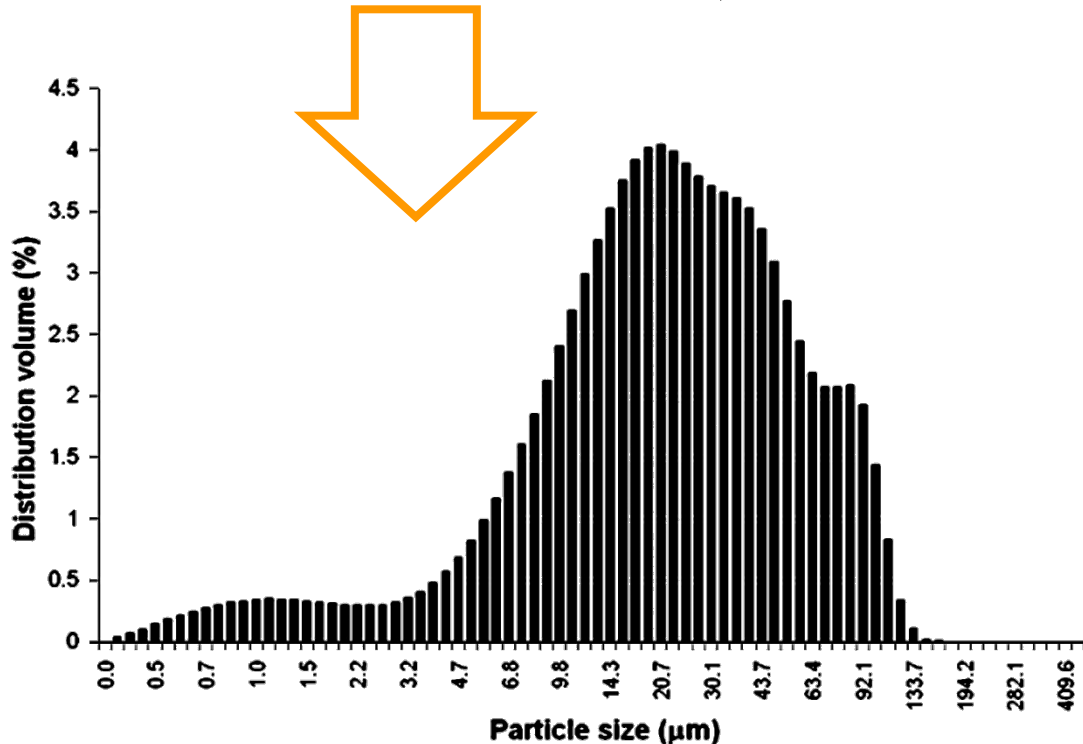
Measures DC resistive pulses for cell **counting and sizing**



Shape of Resistive Pulses



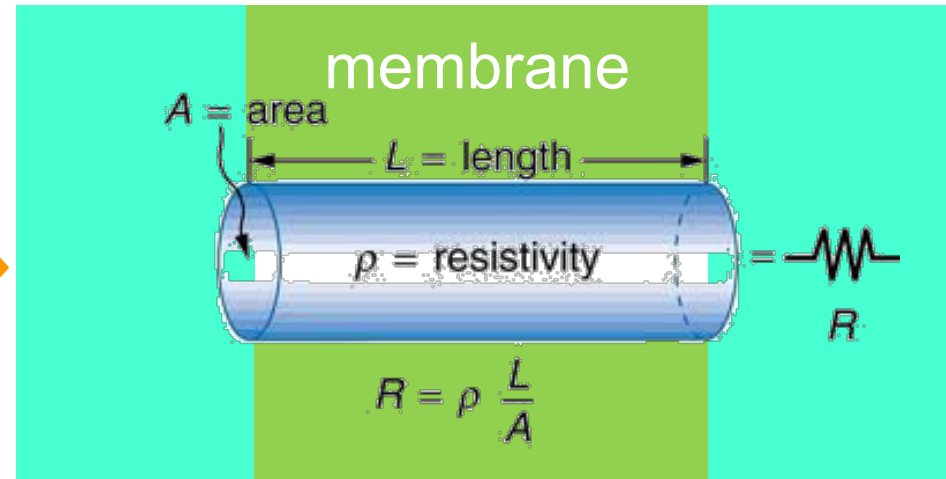
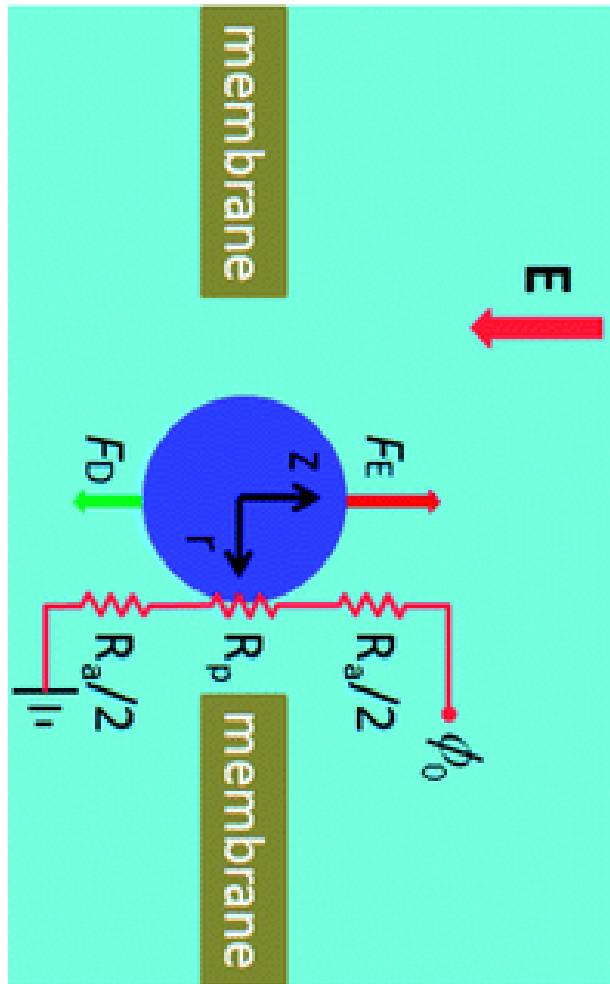
$$\frac{\Delta R}{R} \approx Volume$$





Pore Resistance

For circular holes:

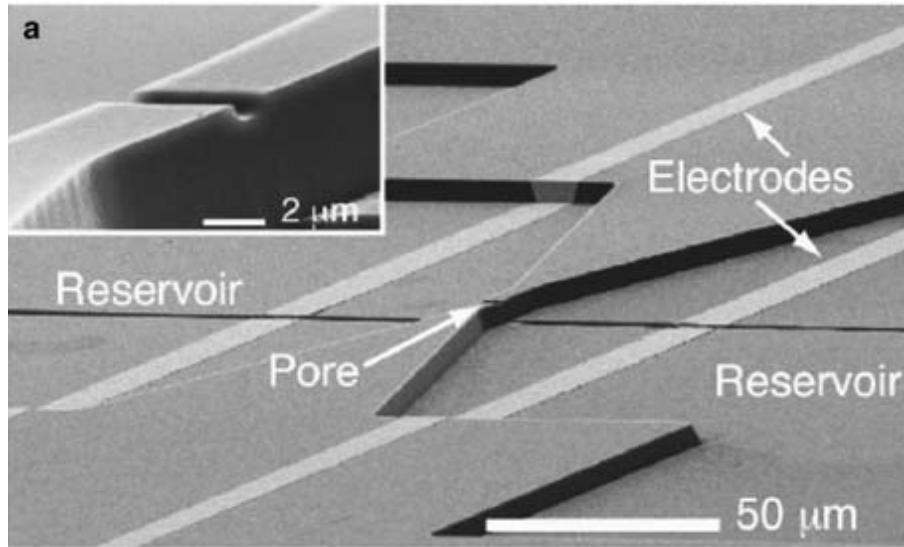


μm	Aperture diameter, μm
0.4 - 12	20
0.6 - 18	30
1.0 - 30	50
1.4 - 42	70
2.0 - 60	100
2.8 - 84	140
4.0 - 120	200



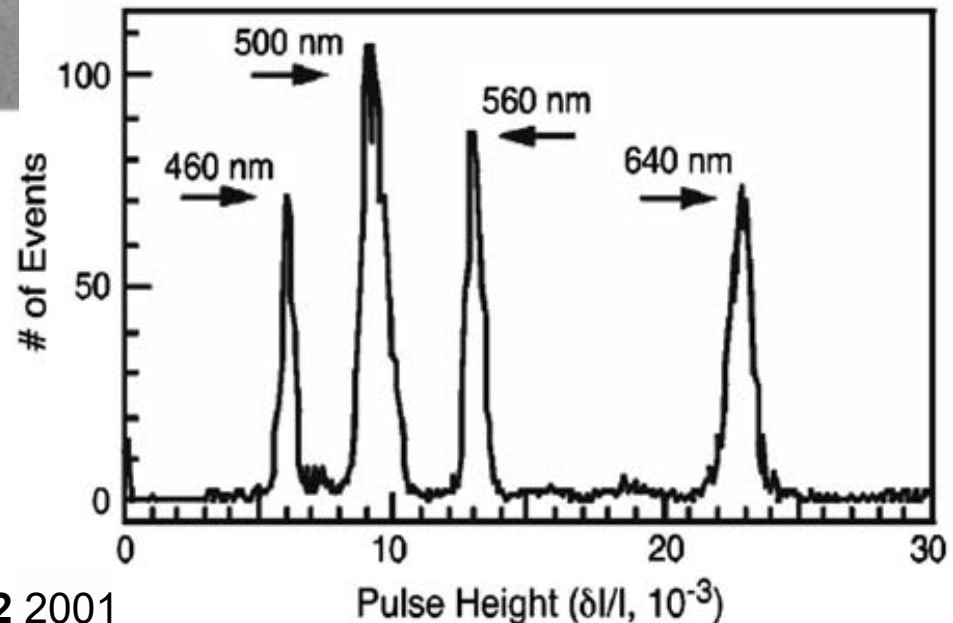
Evolution of Microscale Coulter Counters

MEMS implementations of the Coulter counter:



1 μm aperture

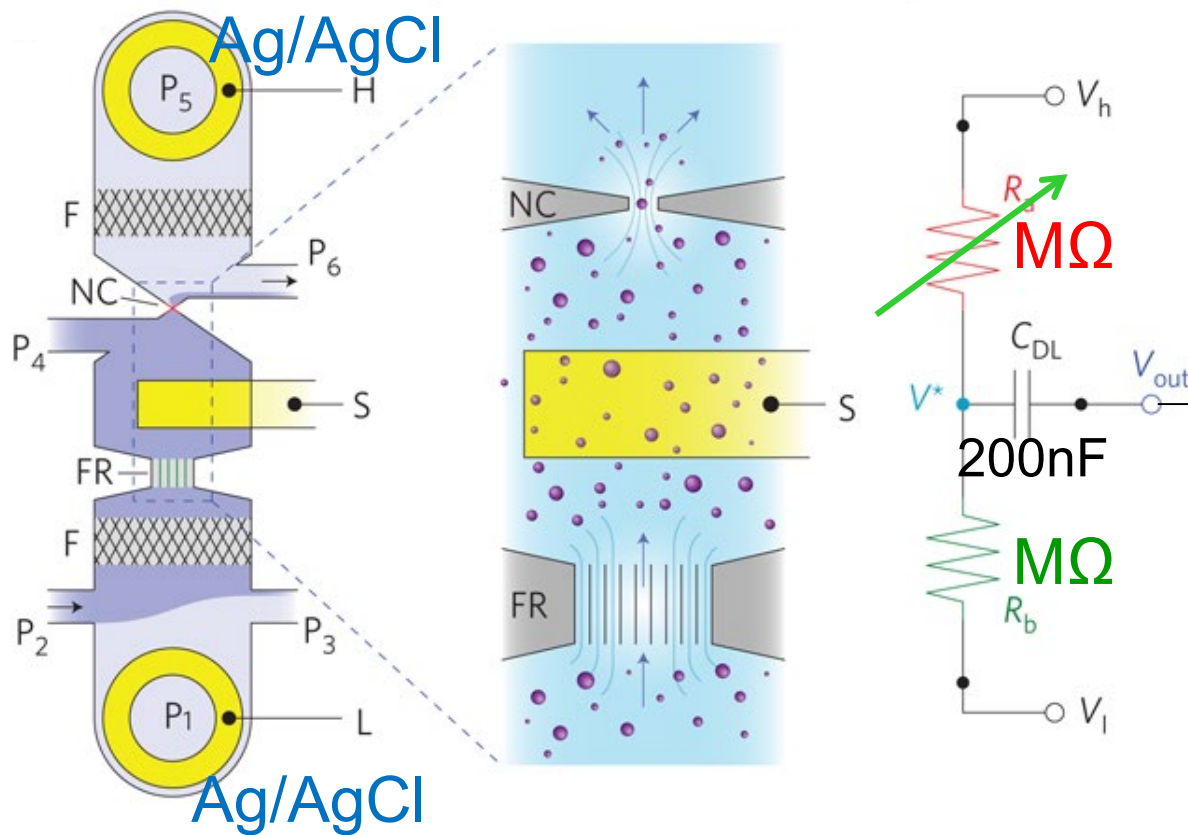
$\pm 10\text{nm}$ size resolution



O. A. Saleh and L. L. Sohn, *Rev. Sci. Instrum.* **72** 2001



Nanoparticle Counter Implementation



ARTICLES

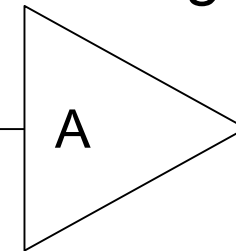
PUBLISHED ONLINE: 6 MARCH 2011 | DOI: 10.1038/NNANO.2011.24

nature
nanotechnology

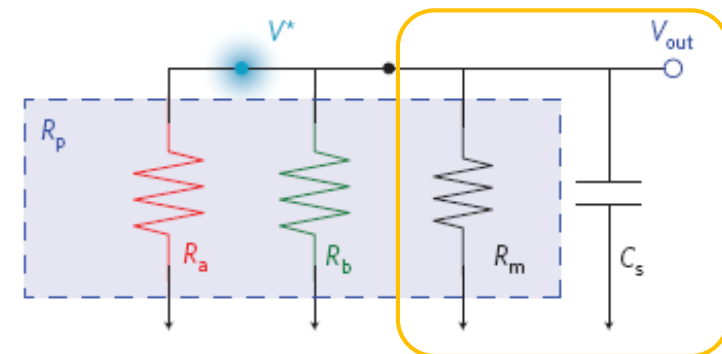
A high-throughput label-free nanoparticle analyser

Jean-Luc Fraikin¹, Tambet Teesalu², Christopher M. McKenney¹, Erkki Ruoslahti^{2,3} and Andrew N. Cleland^{1*}

Voltage reader



Amplifier input
impedance sets the
BW = 650kHz

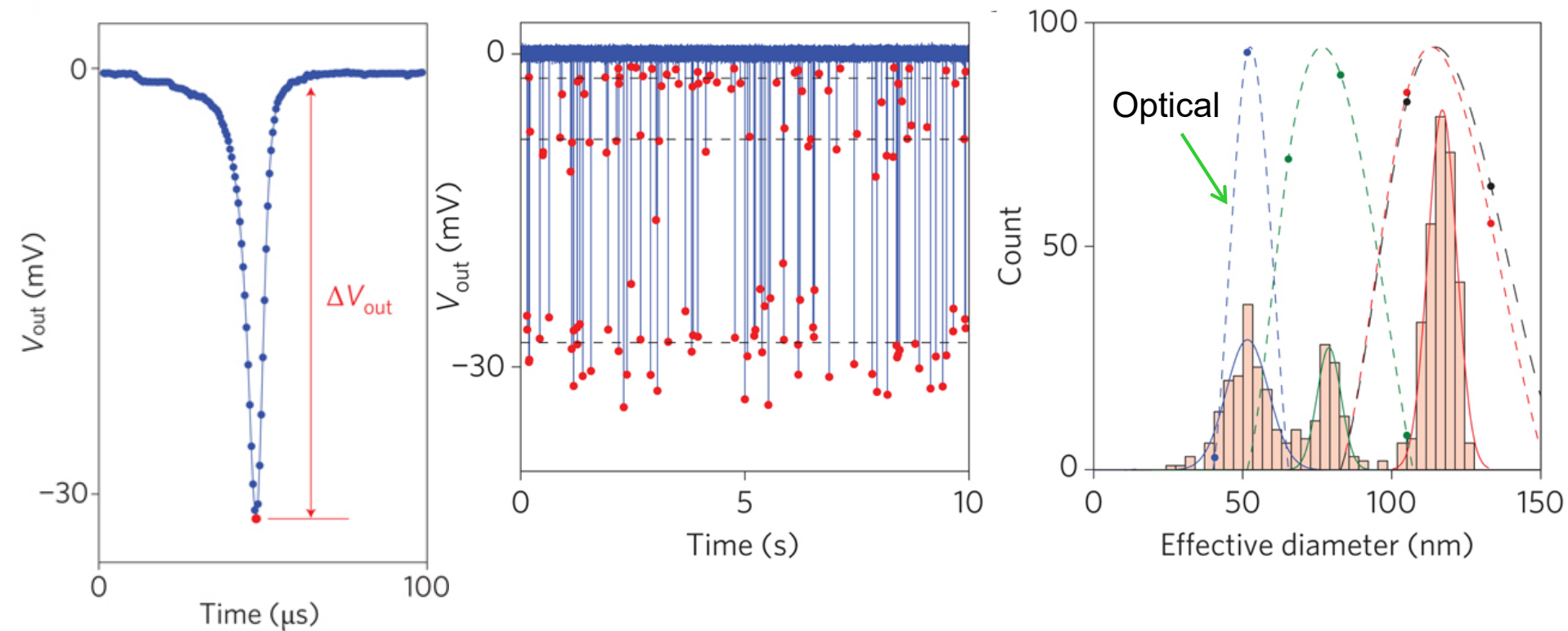


Fluidic resistors for biasing
Capacitive output coupling



High Temporal and Size Resolution

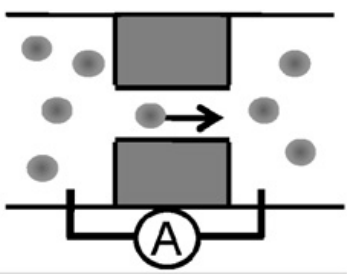
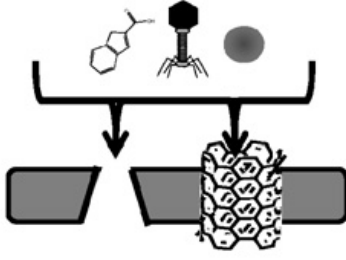
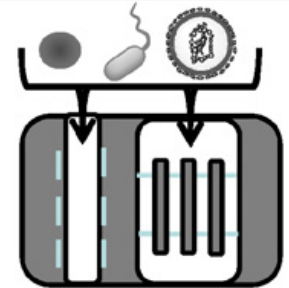
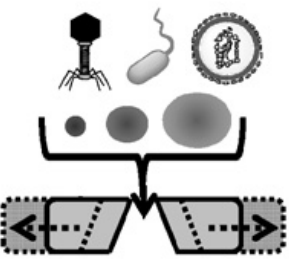
Resolving between 50nm, 75nm and 117nm diameters with a throughput of **500'000 particles/sec**





Miniaturization Trend

Evolution of Resistive Pulse Sensor Development

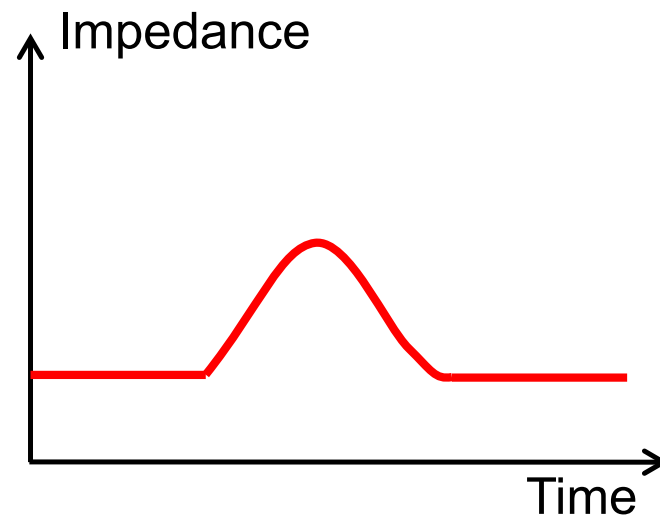
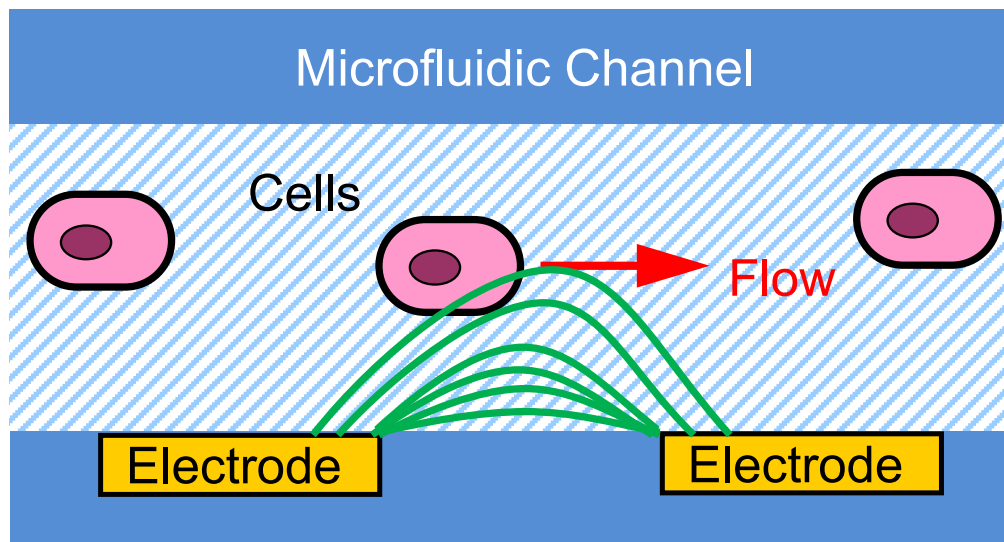
Coulter Counter	Biological Pore	Synthetic Nanopores	Microfluidic Sensors	Elastic Pore
				
Fixed pore (μm)	Fixed pore ($\text{\AA}$)	Fixed pore (nm)	In-series & parallel pores (μm)	Flexible pore (μm to nm)
Large particles > 500nm	ssDNA	Molecules to Nanoparticles	Micro to Nano particles	Micro to Nano particles
Conc, Size & Shape	Molecule Size	Conc, Size & Charge	Conc & Size	Conc, Size & Charge

Pros: highly sensitive (from cells down to molecules)

Cons: difficult fabrication, risk of clogging, only sizing



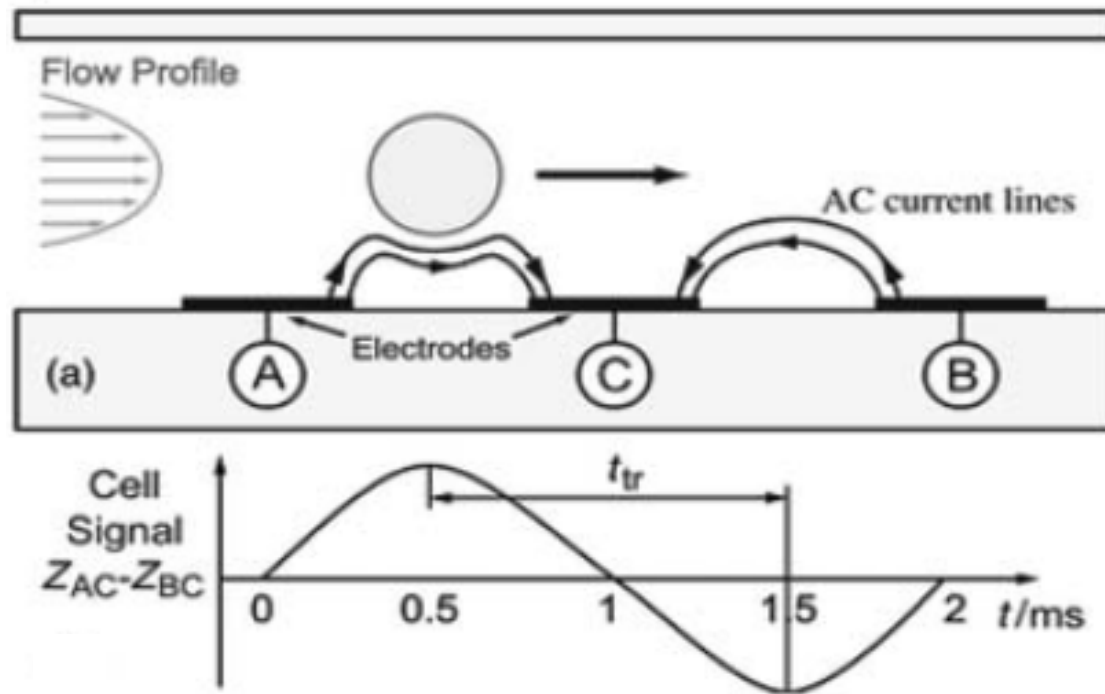
Impedance Flow Cytometry



- A novel approach, pioneered by Morgan and Renaud (~2000)
- Tracking impedance between close electrodes over time
- Measurement in AC (to bypass double layer)
- High-throughput single-cell analysis (beyond sizing)



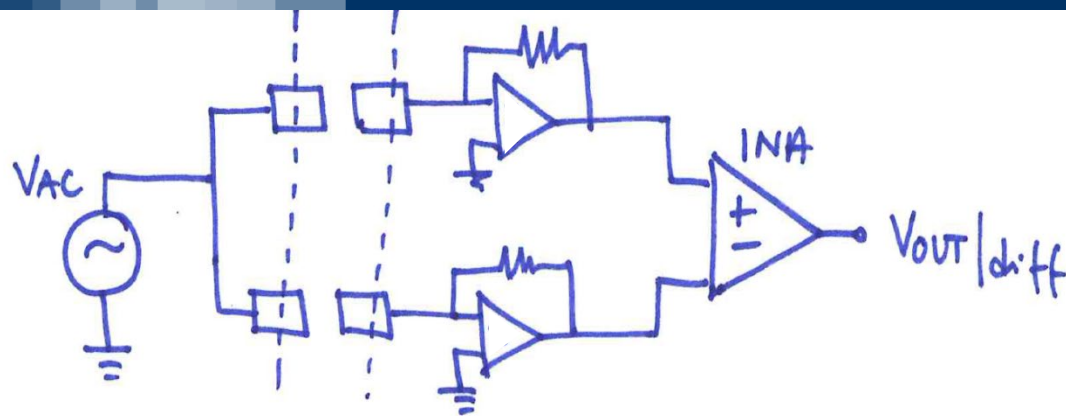
Differential Sensing Configuration



- Three electrodes for differential sensing
- Remove slow drifts
- Bipolar signal
- Speed can be also estimated

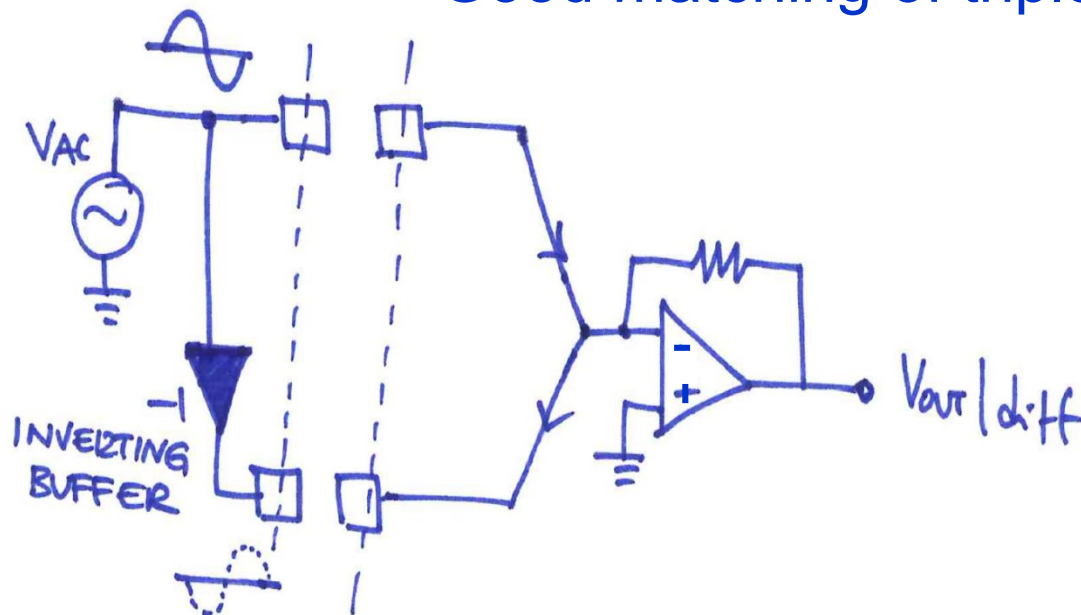


Two Configurations for Differential Sensing



Voltage difference

Good matching of triplet impedances is required

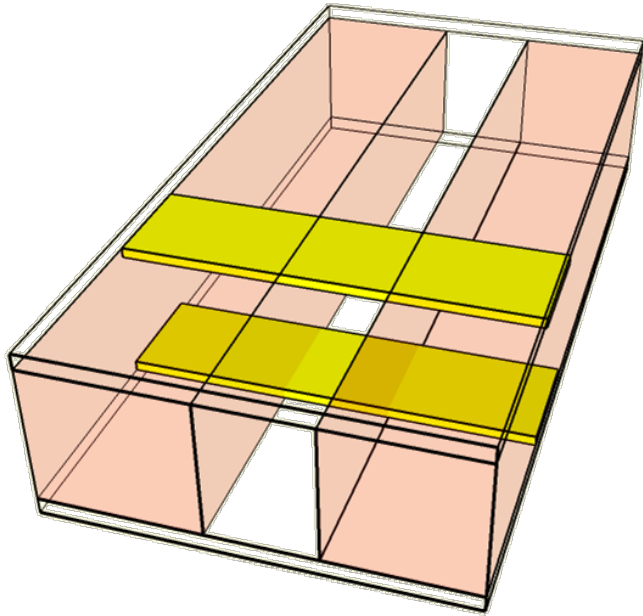


Current difference



Electrodes Configurations

Parallel-plate facing electrodes

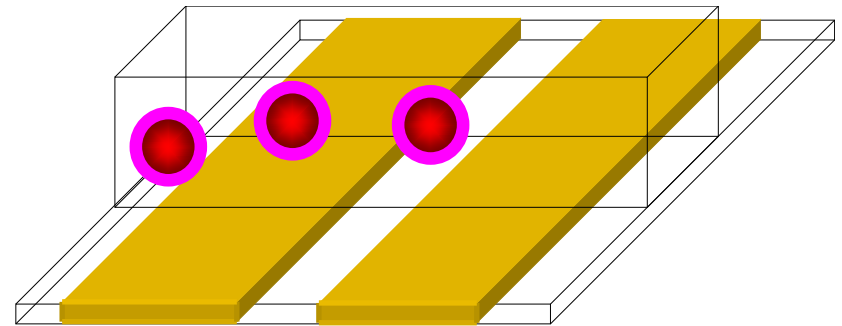


Pros: signal independent of vertical position

Cons: difficult fabrication and alignment, clogging

vs.

Coplanar electrodes



Pros: ease of fabrication, no clogging (if focusing)

Cons: *inhomogeneous field* (vertical dispersion sensitivity, no simple expression)

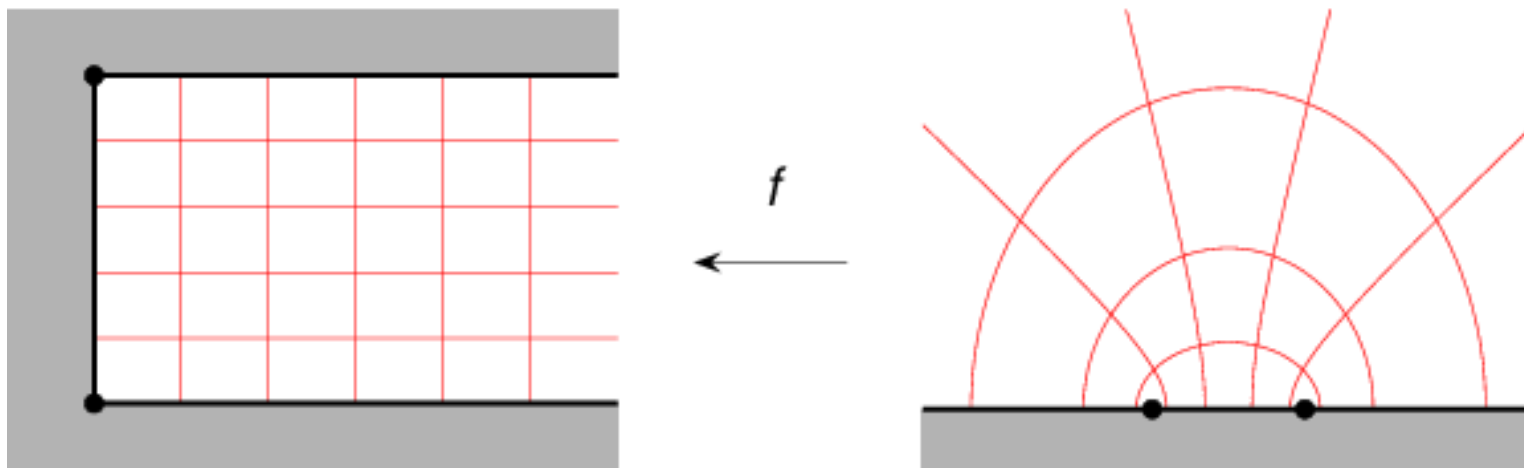


Conformal Mapping

Conformal maps: transformation between holomorphic functions preserving local angles between curves and *harmonicity*:

$$f : U \rightarrow V \quad U, V \subset \mathbb{C}^n$$

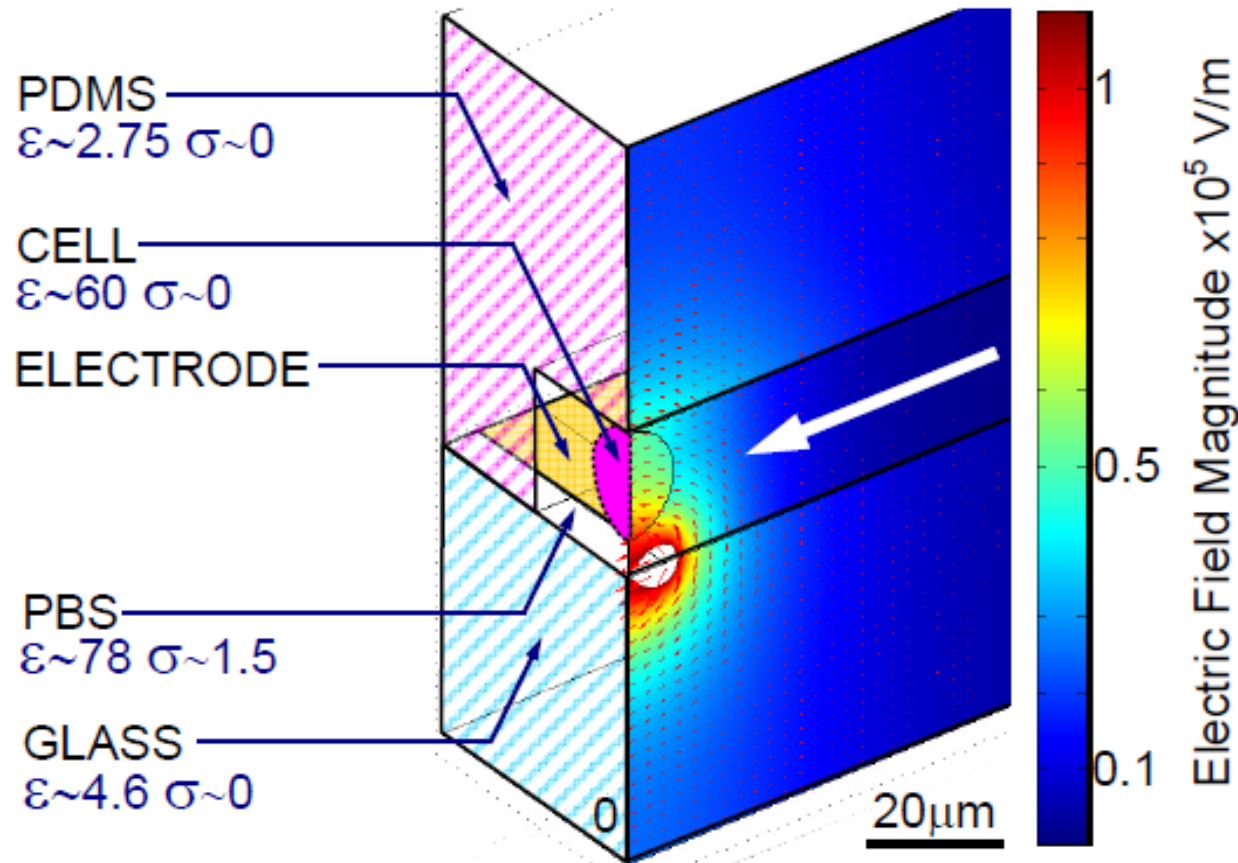
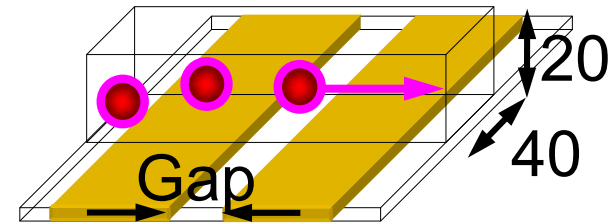
Schwarz-Christoffel transformation maps a semi-infinite plane to a polygon:



Case Study

Design of Sensing Electrodes

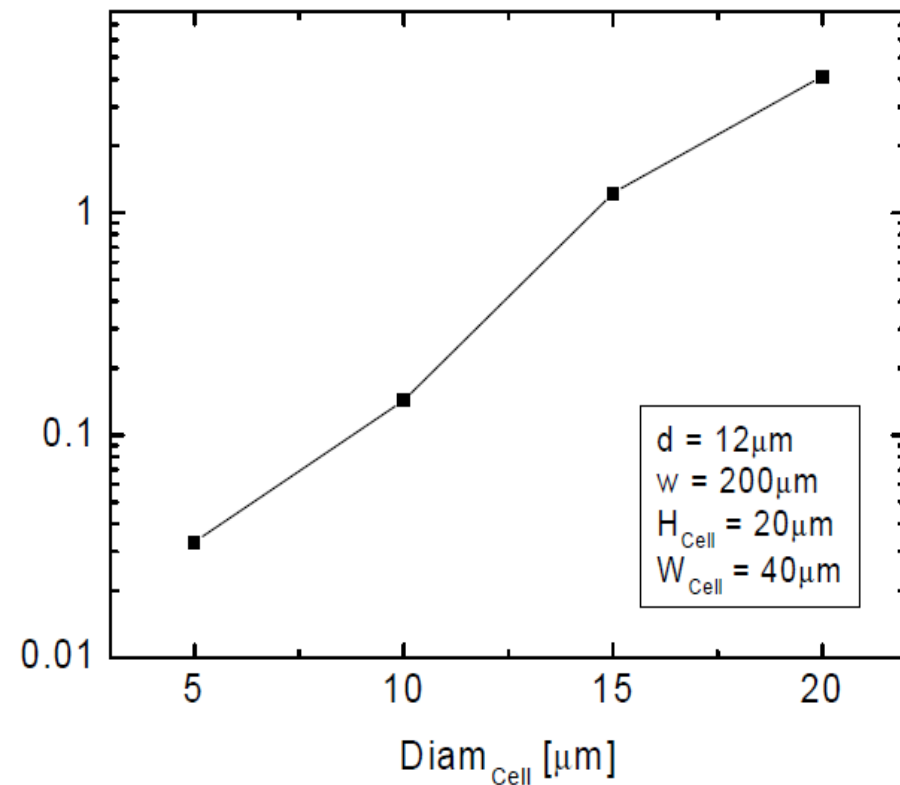
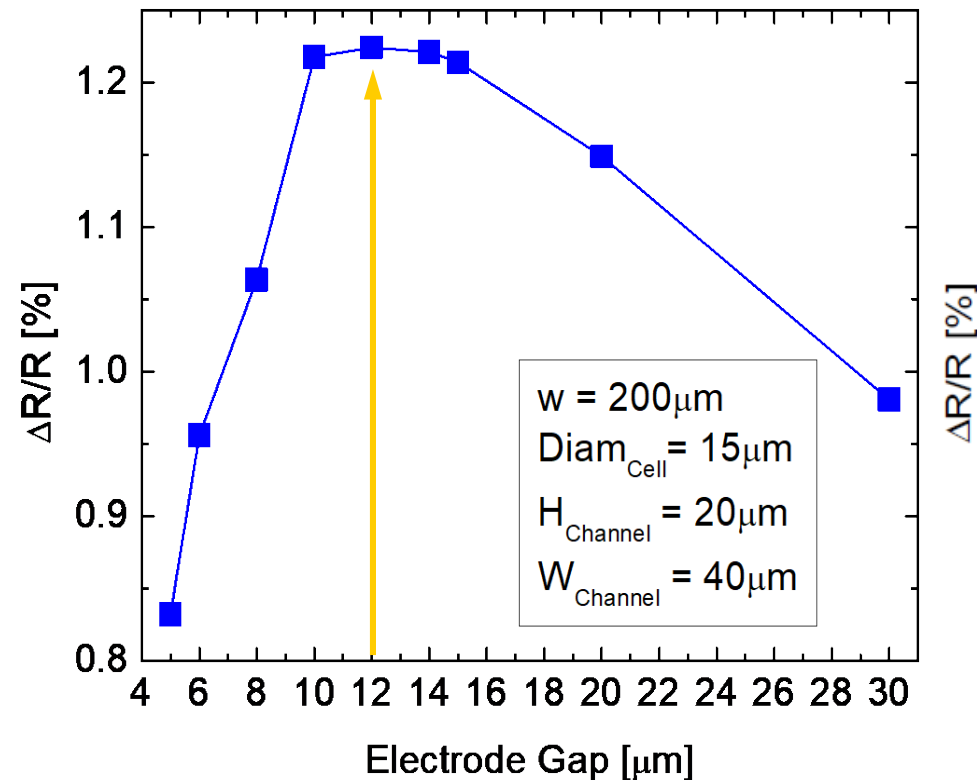
- Simple technology
 - Ease of alignment
 - Target cells 5-15 μm
- coplanar
→ transverse
→ 20 x 40 μm^2





Optimal Electrode Design

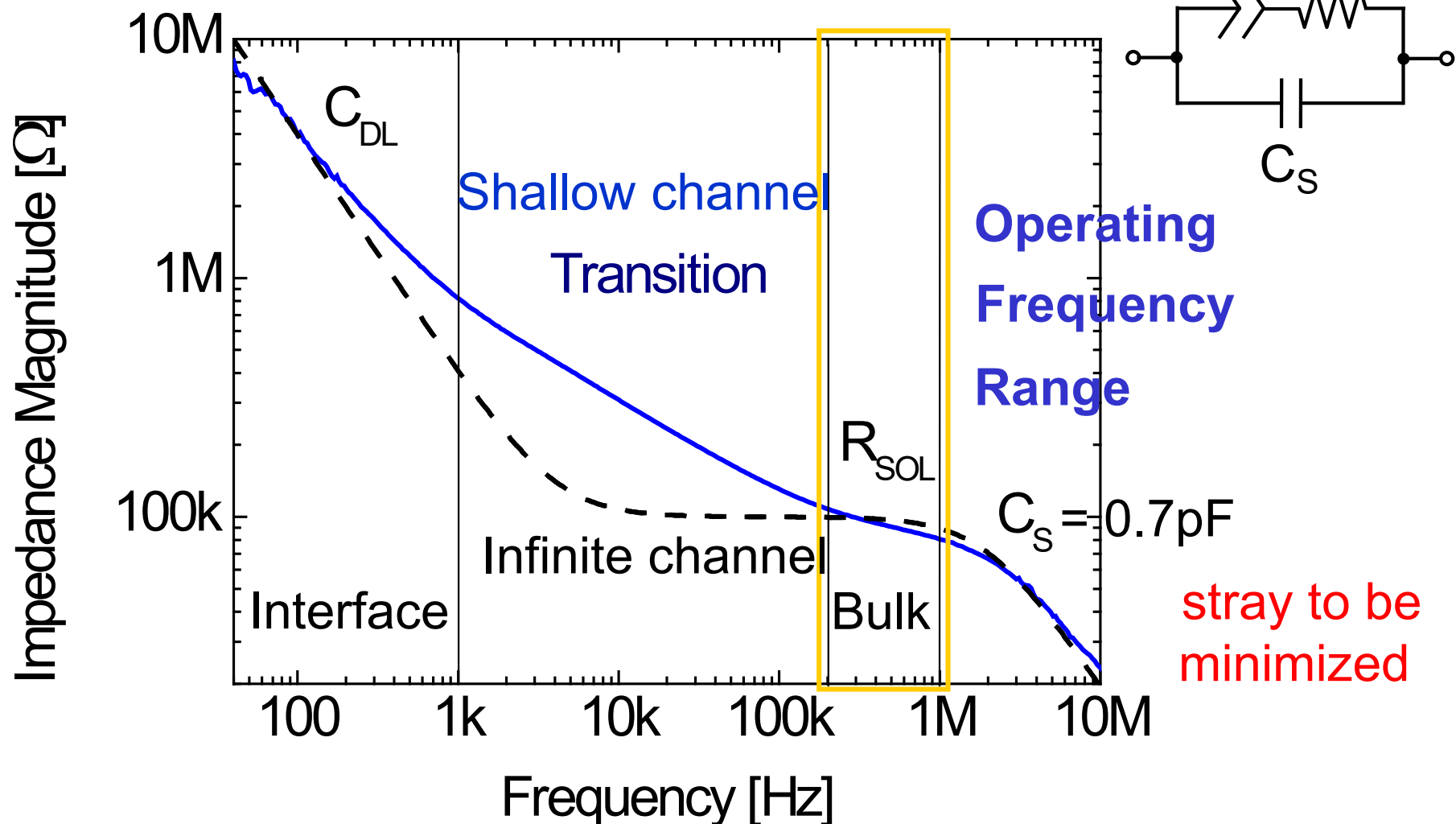
- The gap sets the vertical extension of the sensing volume
- An optimal gap exists for a given cell diameter
- Sensitivity scales with cell volume and vertical height





Operating Frequency

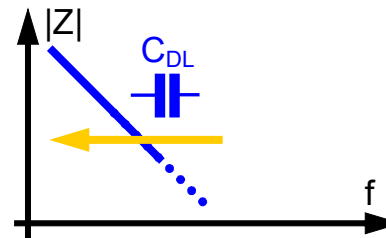
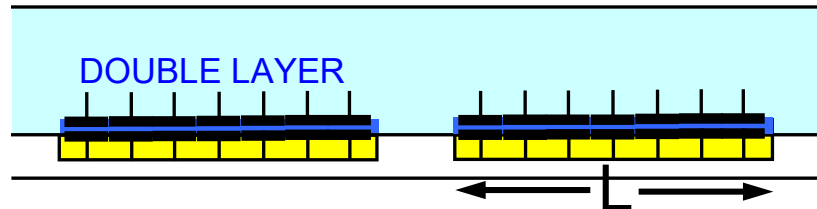
Four regions in the impedance spectrum:



Impact of Geometric Parameters

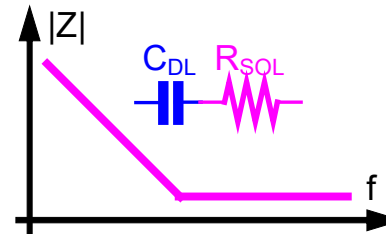
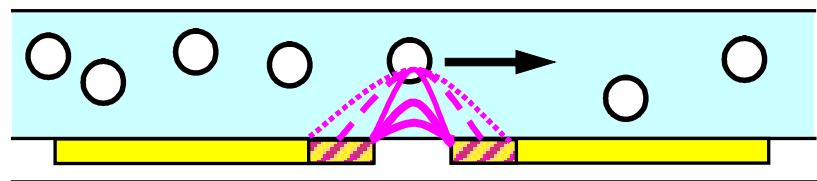
Different geometric parameters control **separate spectral regions**:

LOW FREQUENCY



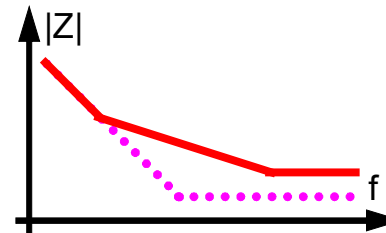
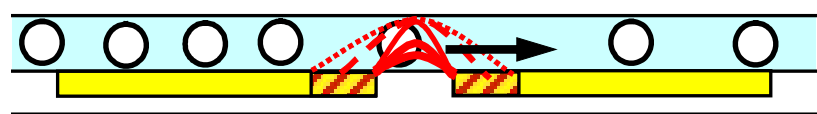
Total area
 $0.1 \text{ pF}/\mu\text{m}^2 \cdot W \cdot L$

MEDIUM FREQUENCY



Fringing fields
Independent of L

SHALLOW CHANNEL

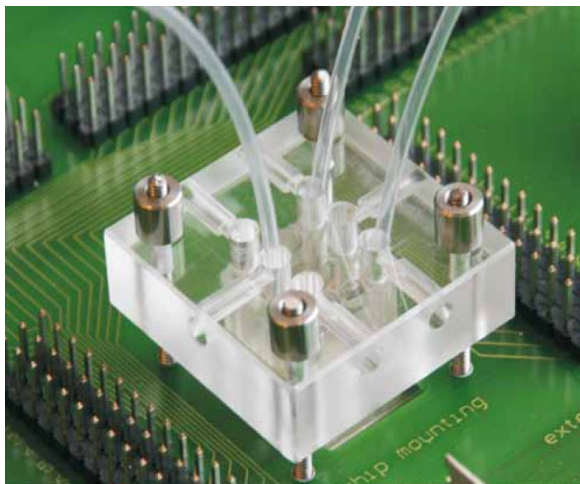


Field compression
Distortion

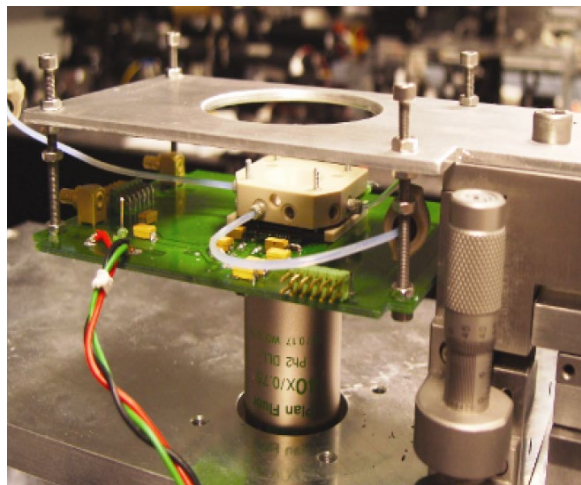
- R_{SOL} set by channel size and gap (max. sensitivity)
- Maximize L (and minimize C_S) to extend the resistive plateau
- Take into account impedance increase due to confinement



Instrumentation: the Need for Miniaturization



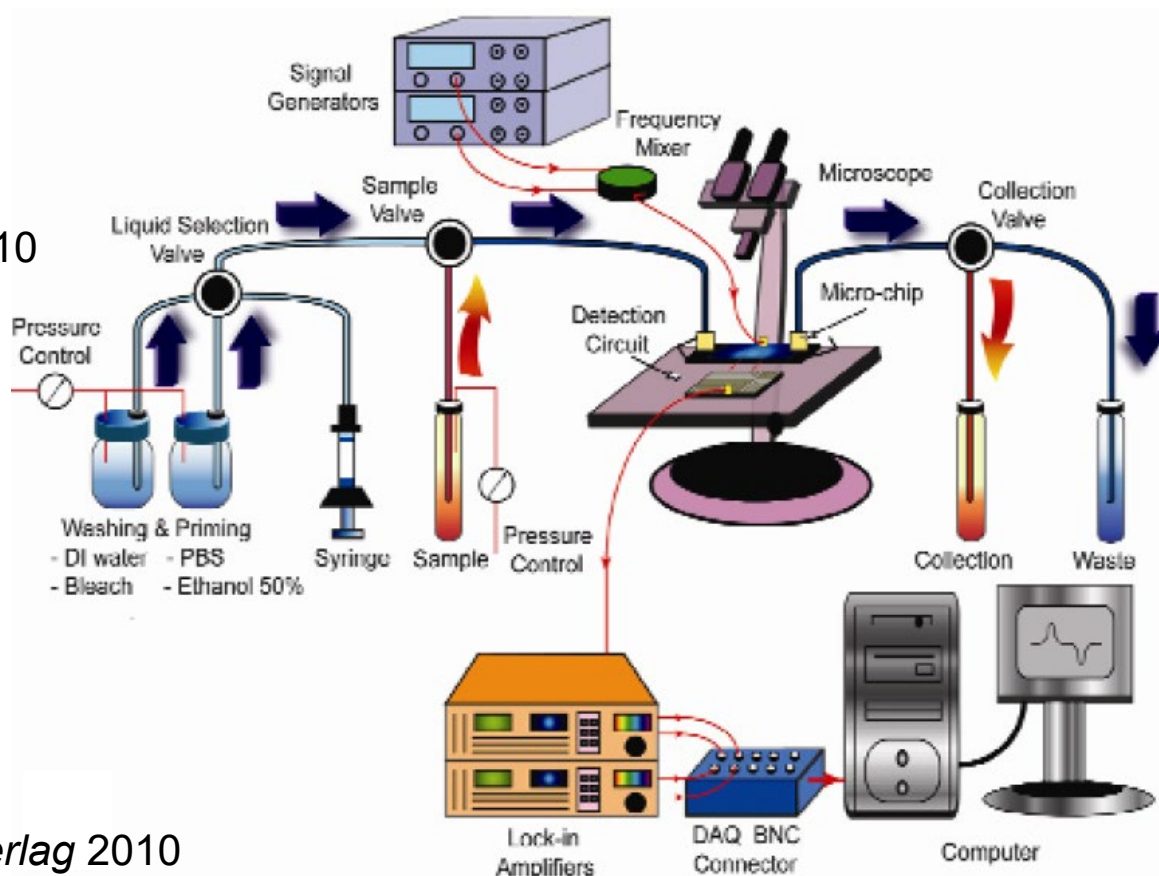
P. Renaud *et al.*, *Lab Chip* 10 2010



T. Sun & H. Morgan, *Springer Verlag* 2010

Show the whole picture...

Not portable!



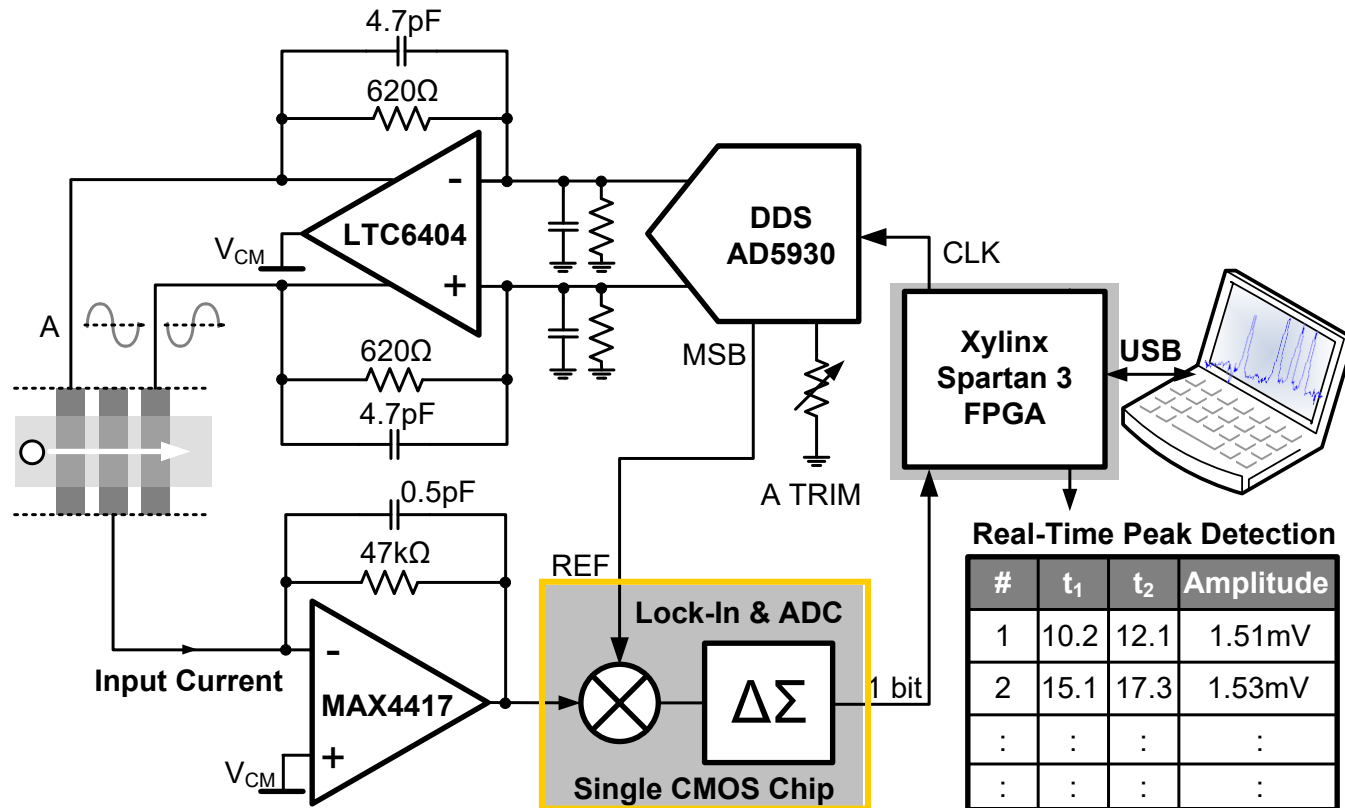
Requirements:

- Operating frequencies: 10kHz – 20MHz
- Amplitudes: 1mV – 1V
- Resistance: 100 Ω – 100k Ω
- Sensitivity: 1% - 0.1% impedance variation
- Sampling rate: >10kSa/s

Issues:

- Clogging of cells in the channel
- Miniaturization
- Avoid size and height variability for discrimination

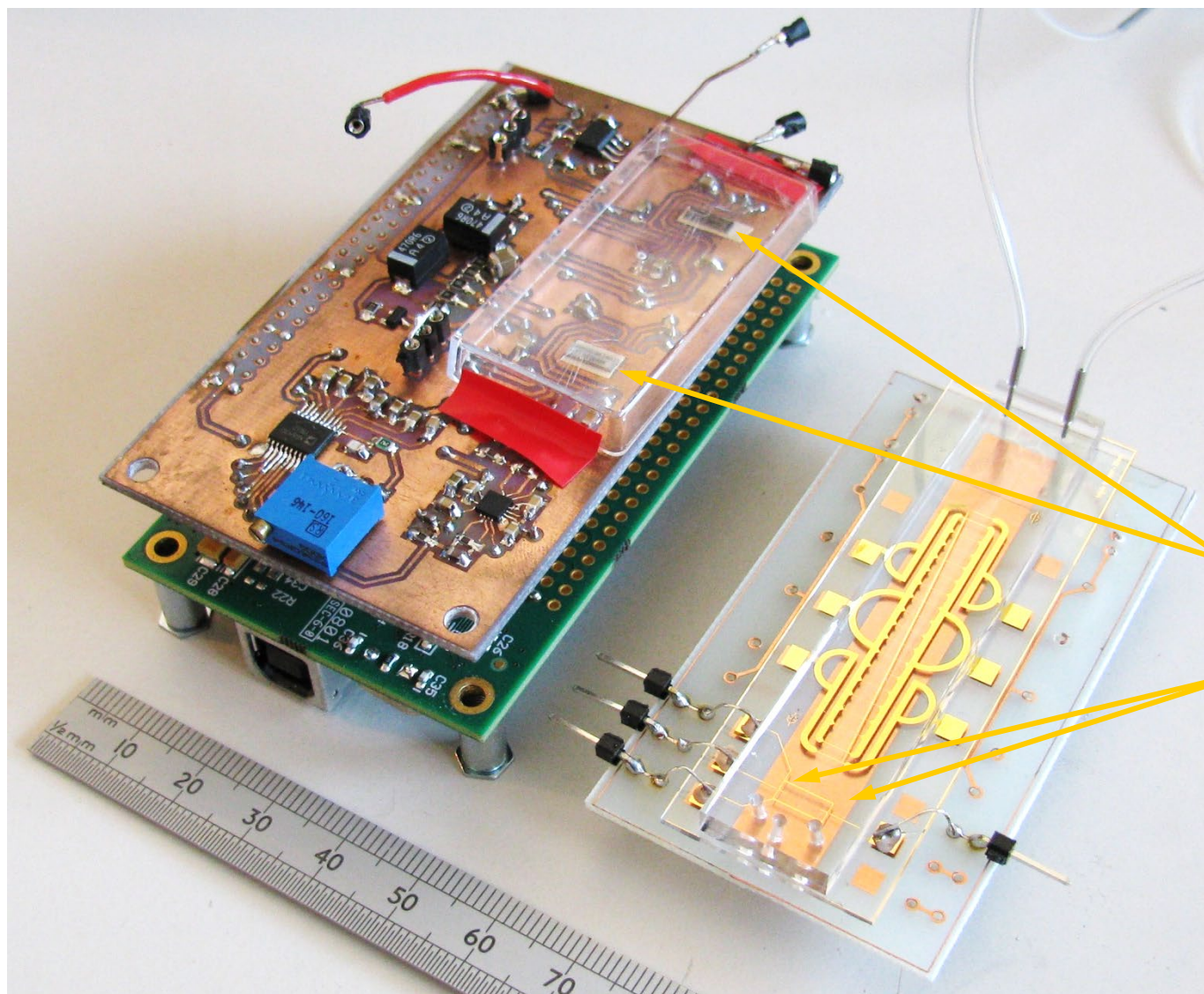
Miniaturized Impedance Detector



- On-board signal generator
- ASIC 1MHz **CMOS lock-in** and 20bit $\Sigma\Delta$ converter
- Real-time FPGA peak detection algorithm
- Digital USB data acquisition



Credit Card Sized Implementation



**Ultra compact
system**

3.3V USB
Power Supply

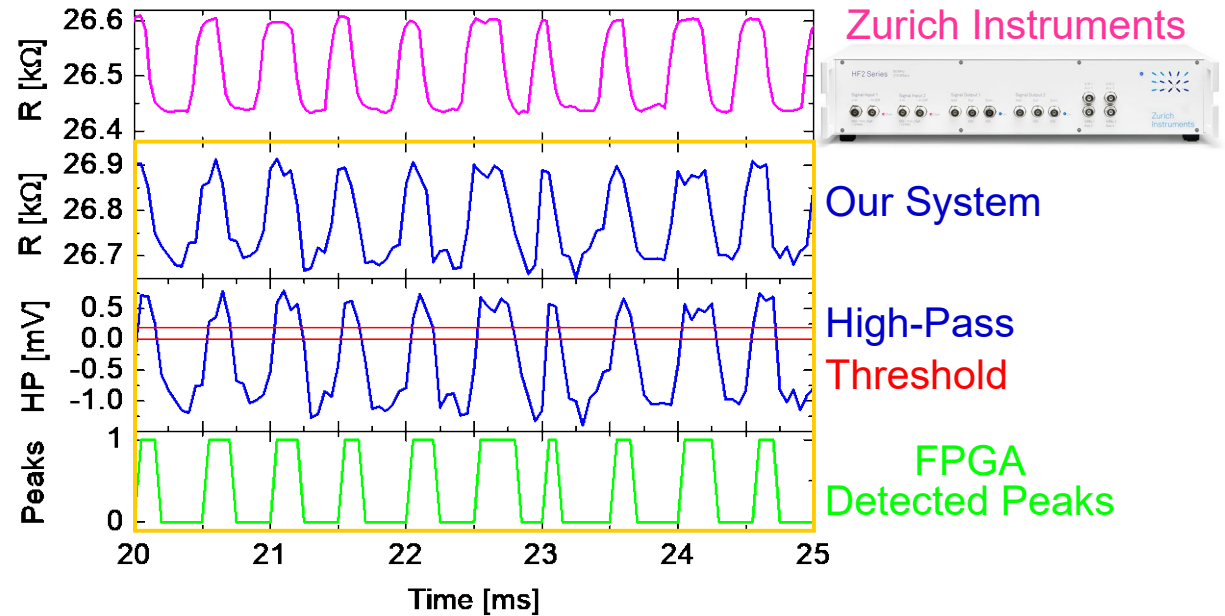
2 channels:

- 2 outlets
- Complex impedance

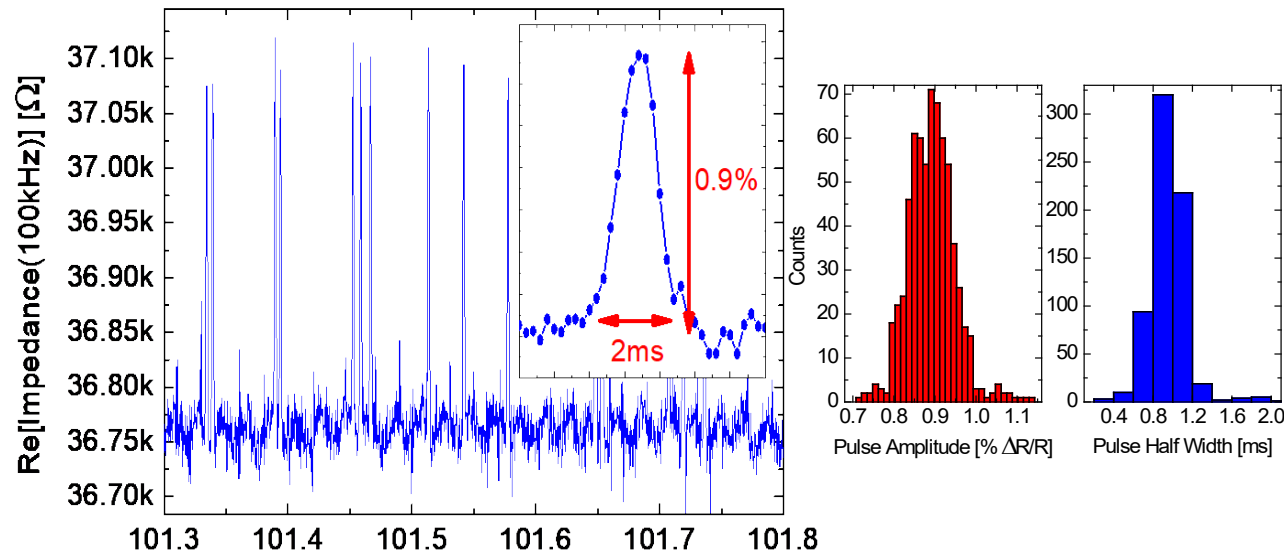


Achievable Performance

2000 counts/sec



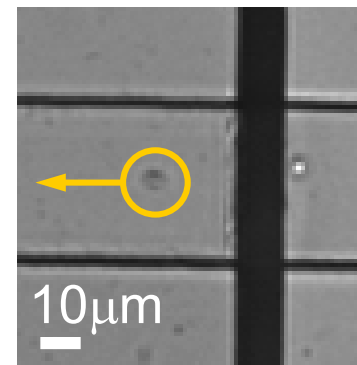
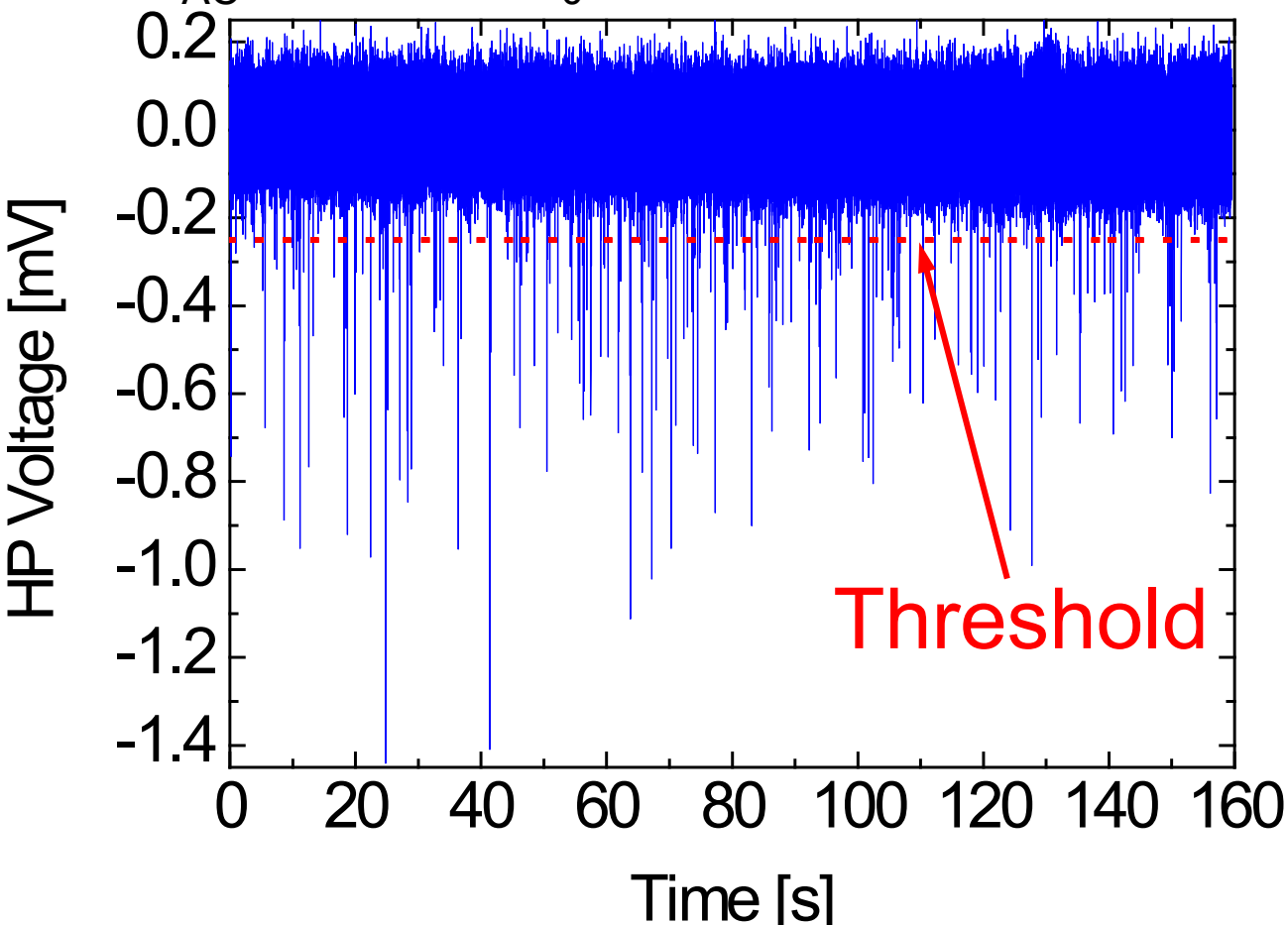
Validation with
10 μ m polystyrene
beads





Counting Yeast Cells

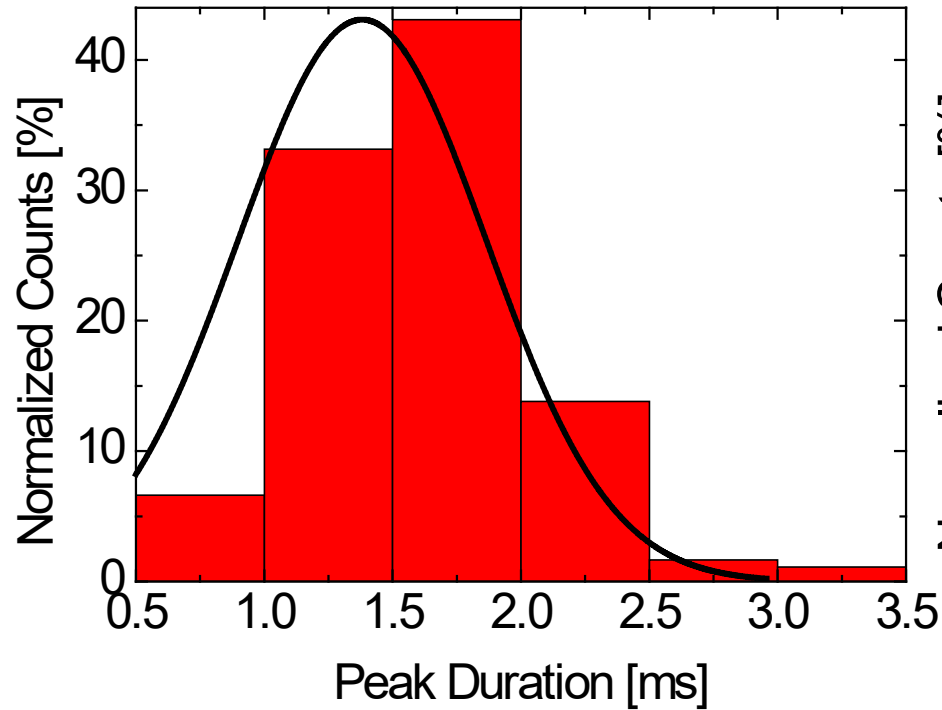
- *Sacc. Cerevisiae* (5 μ m)
- 10⁵ cells/ml in 0.35S/m medium (with BSA)
- $V_{AC} = 650$ mV, $f_0 = 100$ kHz, 2kSa/s



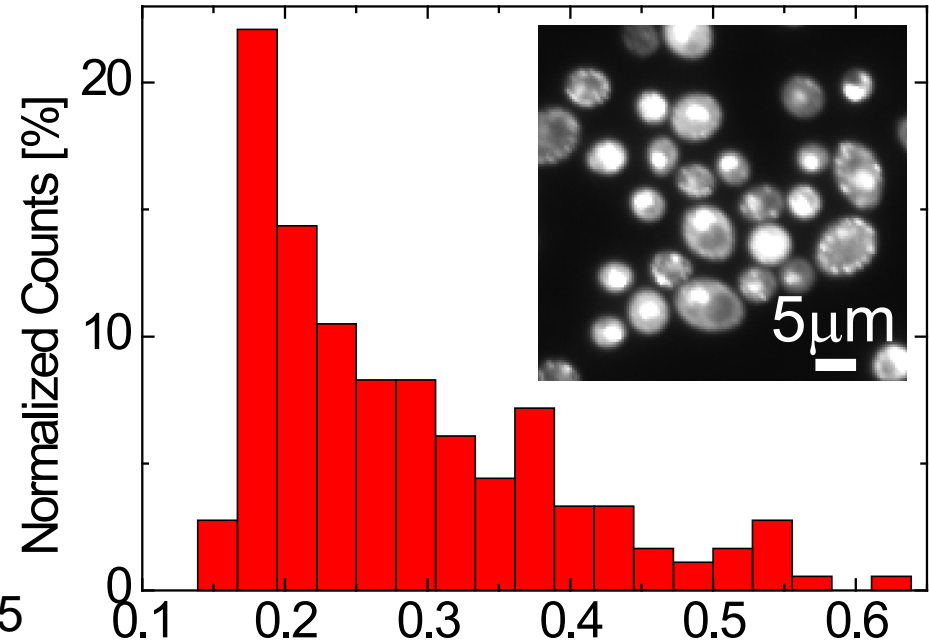


Yeast: Peak Analysis

Pulse Width:



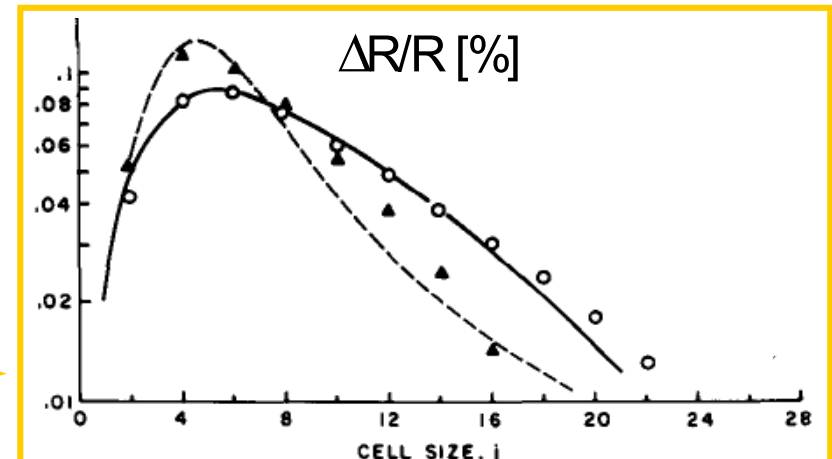
Amplitude:



Consistent with:

- cell velocity (3cm/s)
- cell volume (10 μ m beads/8)
- cell size distribution

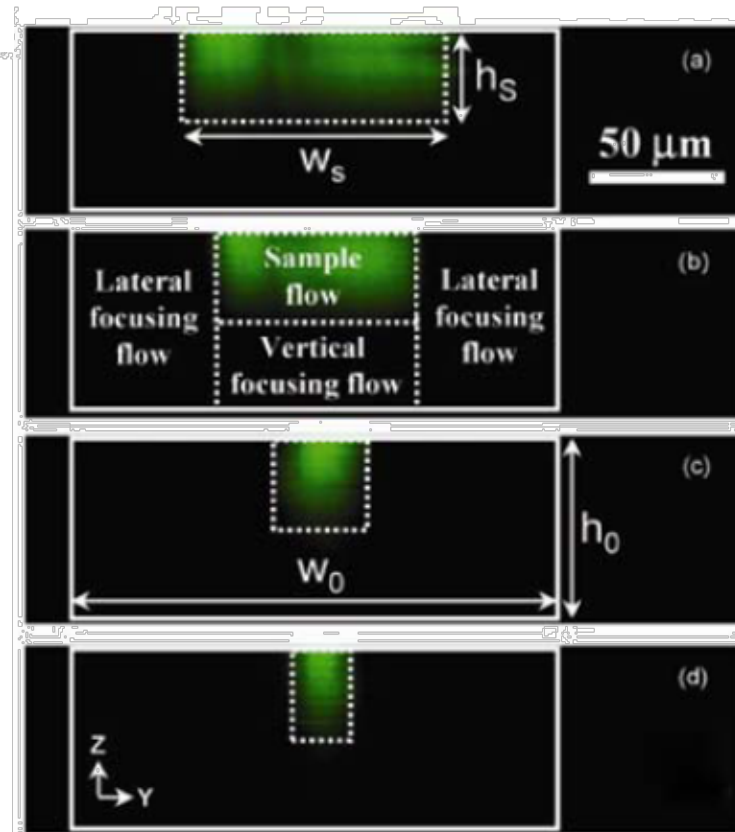
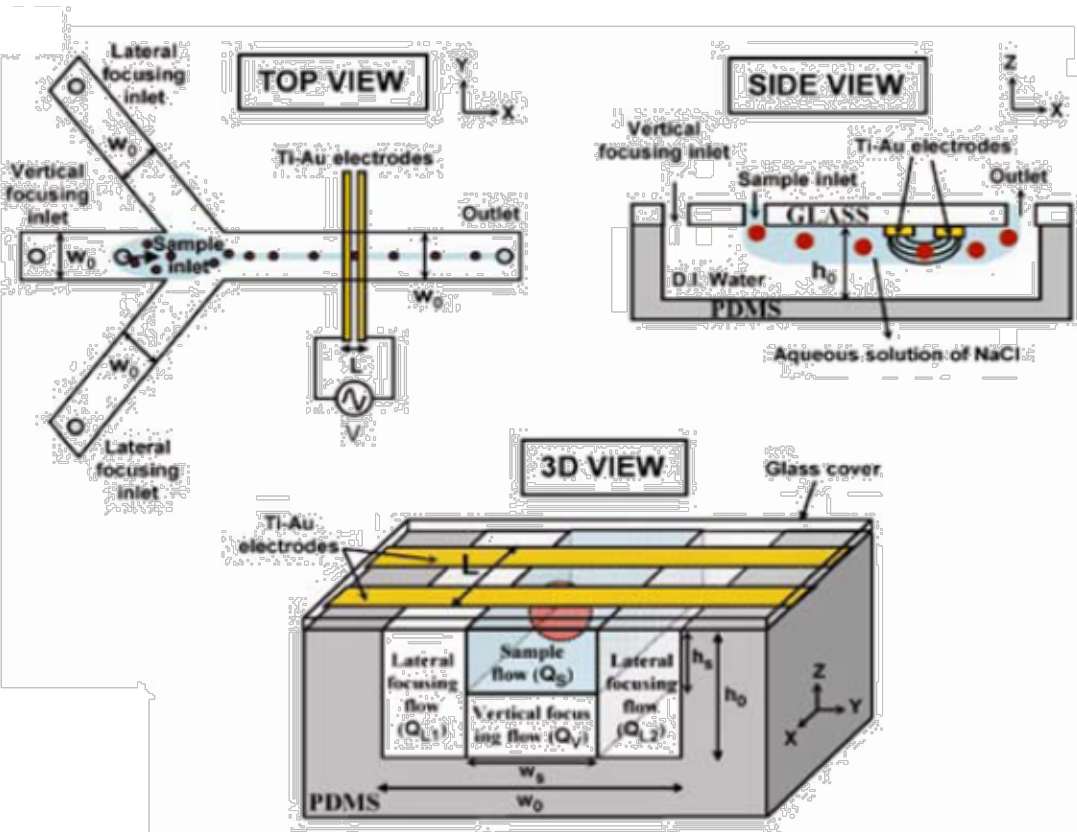
P.Palatt and G. Saidel, *Ann. Biomed. Eng.* 7 1979





Bidimensional Focusing

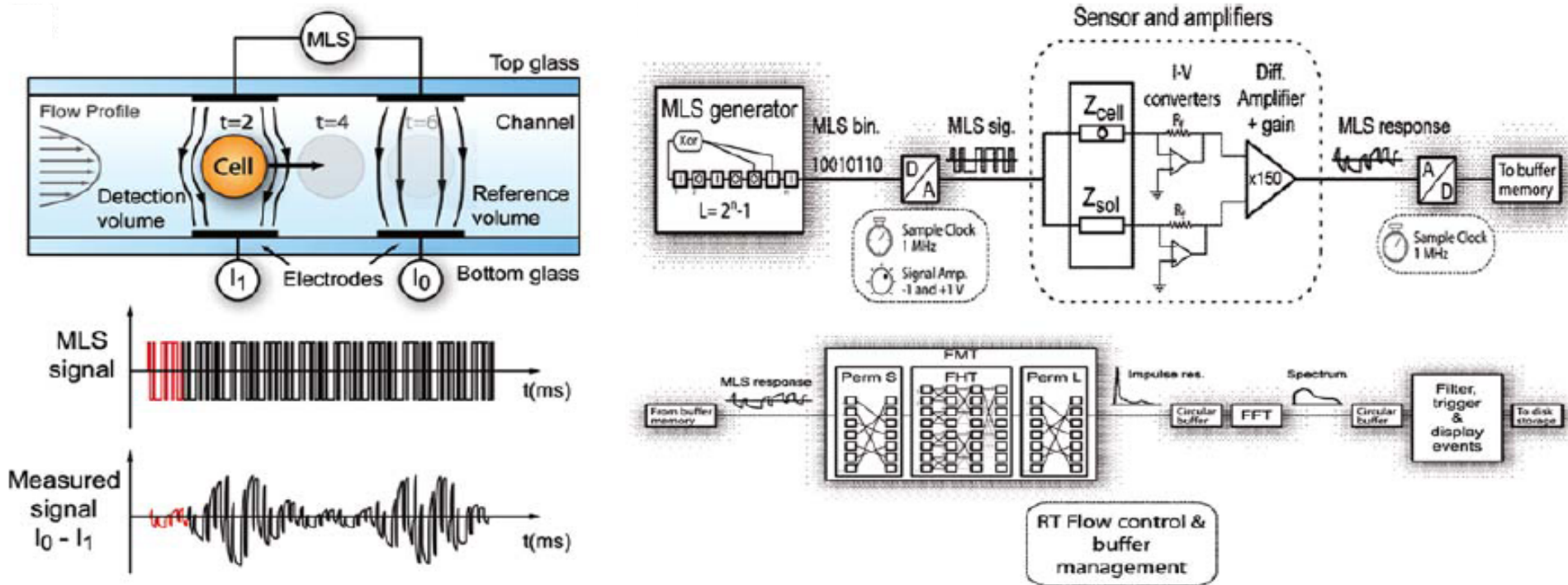
Instead of reducing the cross-section of the channel, **focusing** can be accomplished hydrodynamically or electrically



R. Rodriguez-Trujillo *et al.*, *Biosens. Bioelectron.* 2008

Faster Multifrequency Sensing

Using a pseudo-random noise instead of sinusoid for faster spectroscopy

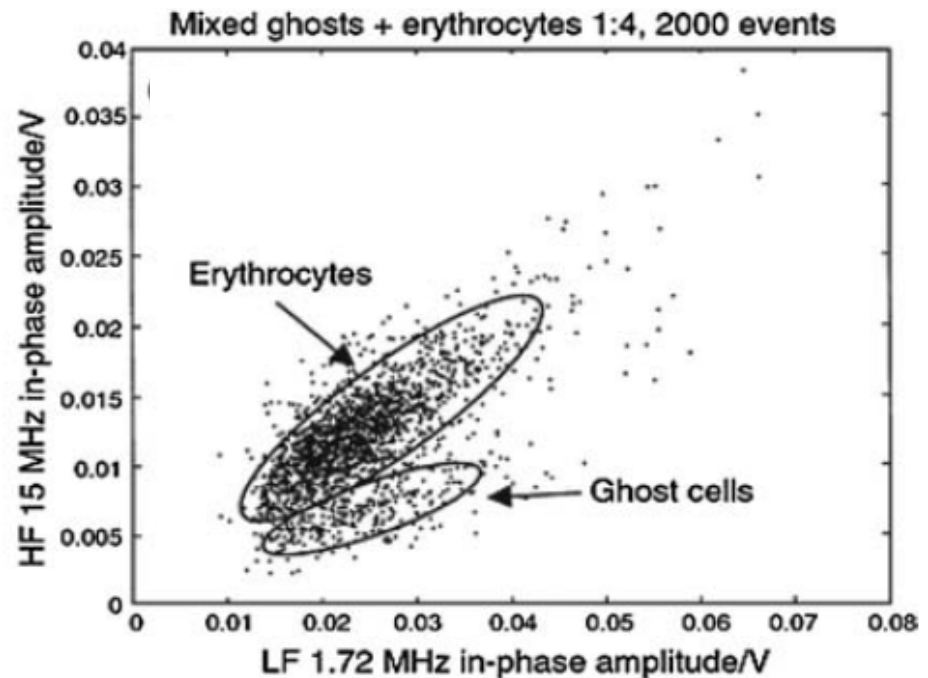
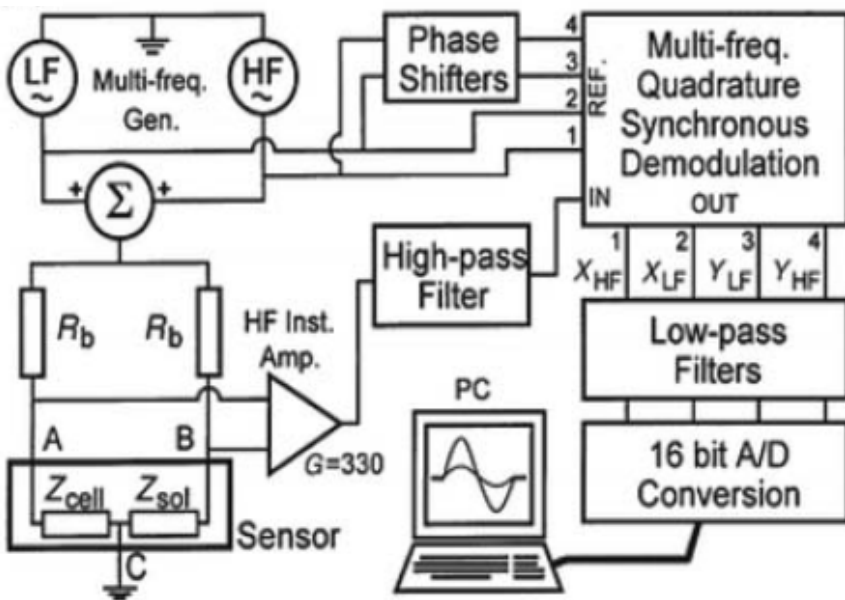


S. Gawad *et al.*, *Rev. Sci. Instrum.* **78** 2007



Intrinsic Cell Sorting

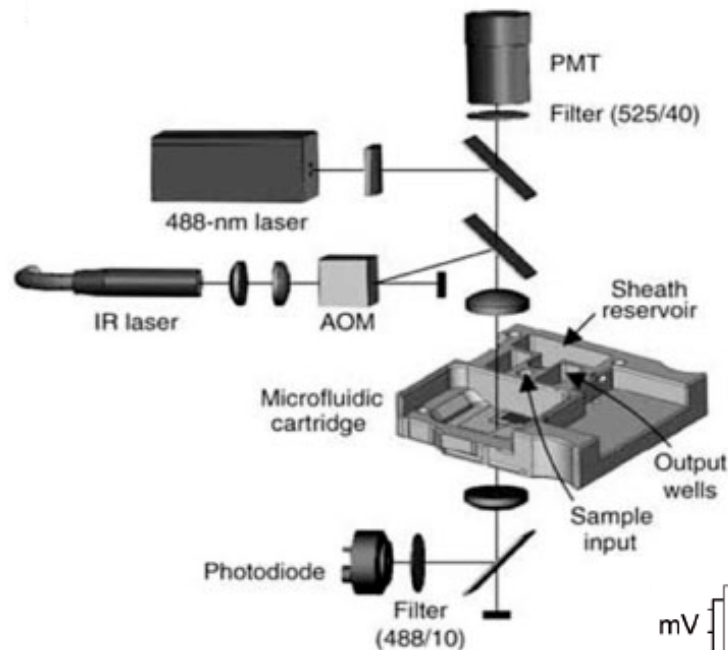
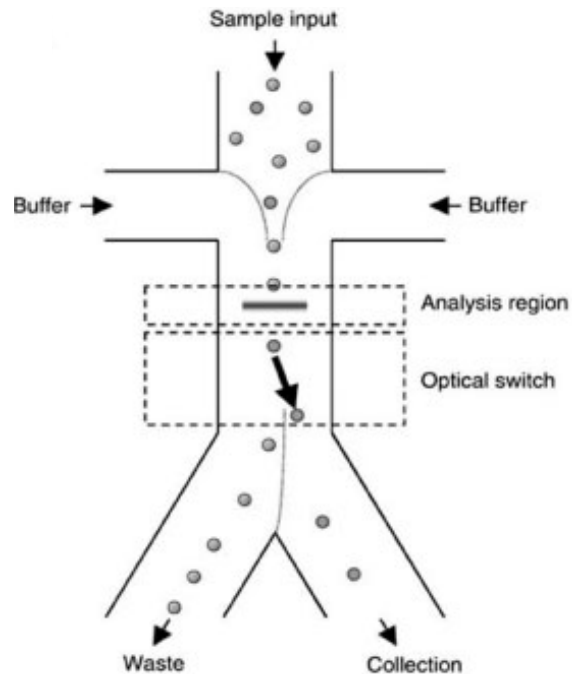
Label-free cell discrimination based on **intrinsic electrical properties**



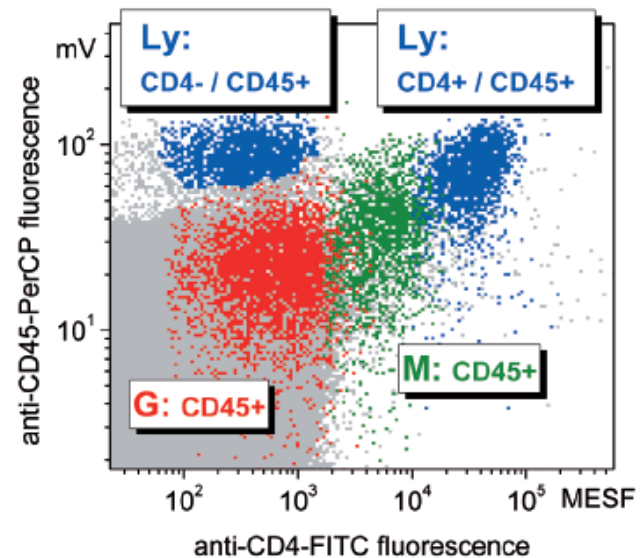
Two frequencies 1.7MHz and 15MHz

S. Gawad *et al.*, *Lab Chip* 1 2001

Cell Sorting: Standard FACS



- Cell separation (purification)
- Relies on highly-specific fluorescent markers
- 1000-10000 cells/sec



A Consolidated Research Field

Authors	Materials Channel (width × height)	Experimental samples	Impedance measurement	Throughput	Summarised results
Morgan et al. (2006, 2007), Benazzi et al. (2007)	Similar to Gawad et al. (2004)	4.5 and 5.5 µm latex beads RBCs Phytoplankton	Differential amplifiers 10 kHz to 10 MHz	100 cells/s	Differentiation of mixed sizes of beads and mixed phytoplanktons
Jagtiani et al. (2006a, b)	Polymer	20 and 40 µm polymethacrylate 20 µm JS pollen and cottonwood pollen particles	Two-point measurement scheme, DC	N/A	A multiple-aperture device
Zhe et al. (2007)	PDMS 100 × 50 µm ²	40 µm PMA particles Juniper pollen	Two-point measurement scheme, DC	N/A	A multiple-channel device
Wood et al. (2007a)	PDMS 100 × 50 µm ²	Barcodes	Network analyser 6.8 MHz	1000 sample/s	Electronic barcode reading technology
Sun et al. (2007d, e), Gawad et al. (2007)	Similar to Gawad et al. (2004)	5.49 and 7.18 µm latex beads RBCs	Trans-impedance 1–500 kHz	100 cells/s	Multi-frequency impedance analysis of single cells in a short time period using MLSs
Jang and Wang (2007)	PDMS 100 × 25 µm ²	HeLa cell	Impedance analyser 1–100 kHz	N/A	Single HeLa cell is captured and measured using impedance
Rodriguez-Trujillo et al. (2007, 2008)	PDMS 190 × 50 µm ² or similar	20 µm beads	Impedance analyser 120 kHz and 1 MHz	20 beads/s	Hydrodynamic focalisation is used to improve sensitivity
Küttel et al. (2007)	Similar to Gawad et al. (2001)	Babesia bovis infected RBCs	Lock-in amplifiers at 8.7 MHz	N/A	Impedance detection of parasitised cell population
Wang et al. (2008)	PDMS 16 × 30 µm ²	CD4+ T cells	MOSFET drain current detection	8 cells/s	Passage of single cells recorded by monitoring variation in MOSFET drain current
Wu et al. (2008a)	PDMS 50 × 16 × 20 µm ³	520 nm, 1, 2 and 4.84 µm beads	Differential amplifier	N/A	Detection of a minimum volume fraction of 0.0004%
Scott et al. (2008)	PDMS 100 × 90 µm ²	20 µm beads	Lock-in amplifier 30– 50 kHz	50 beads/s	A 3D hydrodynamic focussing scheme
Zheng et al. (2008)	Parylene PDMS	5, 8 and 10 µm latex beads RBCs	Impedance analyser 100 Hz to 7 MHz	N/A	Platinum-black electrodes to reduce the EDL effect
Hua and Pennell (2009)	SU8 125 × 17 µm ²	Madin-Darby Canine Kidney (MDCK) epithelial cells	Differential amplifiers 50 Hz	N/A	Single MDCK cell is captured and impedance data to osmotic changes
Malleo et al. (2009)	SU8 Height: 29 µm	HeLa cell	Impedance analyser 100 Hz and 2 MHz	N/A	Differential captured single-cell impedance measurements
Watkins et al. (2009)	PDMS 200 × 32 µm ²	CD4+ T-lymphocytes	Lock-in amplifiers 50 kHz	N/A	A 3-D hydrodynamic focussing scheme
Holmes et al. (2009)	Similar to Gawad et al. (2004)	Whole blood	Differential amplifier 500 kHz to 30 MHz	100 cells/s	Three-part differentiation of leukocytes: T-lymphocytes, monocytes and neutrophils in human blood